JOSHUA BUCASAN

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EDUCATION

Nueva Vizcaya State University **Nueva Vizcaya, Philippines**
Master of Science in Forestry **January 2026 - present**
Relevant Coursework: GIS Application in Natural Resource Management

Benguet State University **Benguet, Philippines**
Bachelor of Science in Forestry **June 2019 – June 2024**
Relevant Coursework: Geospatial Methods in Forestry

EXPERIENCE

Quirino State University **Quirino, Philippines**
GIS and Forestry Instructor (Contract of Service) **January – December 2025**

- Instruct 50+ undergraduate students per semester in GIS fundamentals, spatial analysis, and remote sensing through 2 hours of lecture and 3 hours of laboratory work weekly.
- Curated and processed shapefiles and raster datasets from open-source repositories with the aid of Python scripting and AI to support hands-on learning in spatial analysis, digitizing and map layout design.
- Designed and implemented a comprehensive lab curriculum including 12 structured exercises, 12 instructional handouts, and 3 practical exams to ensure student competency.
- Developed and maintained a centralized departmental GIS database, optimizing access to instructional data reducing student project preparation time.

Department of Environment and Natural Resources – CAR, Surveys and Mapping Division **Baguio City, Philippines**
GIS Specialist (Contract of Service) **October – December 2024**

- Validated high-volume survey data daily against approved actual documents, including 30 Survey Plan Information (SPI) records, 200 lot parcels, and 300 scanned documents to ensure data accuracy.
- Collaborated with geodetic engineers to identify spatial discrepancies (gaps, overlaps, projection errors) resulting from data encoding, ensuring data integrity for official records.
- Documented and recommended corrective actions for data inconsistencies, contributing to the quality and accuracy of databases.
- Produced final map layouts for official reports and presentations, translating complex spatial data into accessible visual formats for stakeholders.

SKILLS

- **GIS Software and Tools** - ArcGIS Desktop, ArcGIS Pro, QGIS
- **Remote Sensing** - Satellite imagery interpretation, aerial photo analysis
- **Cartography** - Map design principles, symbolization and layout
- **Data Analysis** - Python (scripting, API integration and automation), Microsoft Excel, basic statistics
- **Database Management:** PostgreSQL and PostGIS

LICENSE

- **Registered Professional Forester** **October 2024**

PROFESSIONAL TRAINING & DEVELOPMENT

- Facilitator: QGIS Applications in Agriculture (Phase I) - Agricultural Training Institute - Cordillera Administrative Region (40 hours)
- Facilitator: Introductory Training on Geographic Information System and Unmanned Aerial Vehicle Application in Mapping - Benguet State University Center for Geoinformatics (36 hours)
- Training on Remote Sensing 101: Applications in Agriculture, Forestry, and Natural Resource Management - Benguet State University (24 hours)